

fier, MAX40108 precision operational amplifier, and MAX31343 real-time clock (RTC) with integrated MEMS oscillator. The MAX41400 enhances sensor system accuracy by 4x and extends battery life by 55 percent compared to the closest competitor. It provides low offset of 25µV, low noise, and programmable gain with only 65µA current consumption. The MAX40108 requires supplies as low as 0.9V, which helps double sensors' battery life. The MAX31343 provides timekeeping accuracy of ± 5 ppm, plus robust protection by its integrated resonator that eliminates crystal mechanical failures and enables the smallest WLP compared to any other in the market.



Maxim Integrated
www.maximintegrated.com

Snap-in electrolytic capacitors

EPCOS aluminium electrolytic capacitor series B43548 with snap-in terminals claim excellent maximum ripple current capability of 9.80A (400V, 100Hz, 60°C). This makes them ideal for demanding applications in frequency converters, industrial power supplies, UPS systems, medical devices, and photovoltaic inverters. The capacitors are designed for a maximum operating temperature of 105°C, have a rated voltage range of 400V to 500V, and reach under these conditions a useful life-time of 3,000 hours. Depending on the rated voltage, they offer a capacitance range of 68 to 820µF in case sizes of 25mm × 25mm to 35mm × 55mm (d × l). Their high level of resilience against rapid charging and discharging cycles helps them meet the demanding requirements of servo drive applications as well.

TDK Corporation
www.tdk-electronics.tdk.com

Low-power cellular IoT chipset

Altair ALT1255, a new low-power NB2 chipset designed with an integrated SIM (iSIM), user MCU, rich application layer, and GSM/GPRS fallback modem is 5G ready. It will enable low-power, cost-sensitive connected devices to monitor, manage, and control critical infrastructure, medical devices, logistic trackers, and a large variety of LPWA applications. The ALT1255, developed in-house by Sony Semiconductor Israel, shares software architecture, modem application and networking layer, and APIs for fast and easy integration within the Altair product family.

Altair Semiconductor
www.altair-semi.com



AI-enabled gas sensor

Bosch Sensortec's BME688 is the world's smallest four-in-one air-quality MEMS sensor, which combines gas, humidity, temperature, and barometric pressure sensing with innovative AI capability.



It reduces total cost of ownership, cuts development time, and simplifies design. BME688 is ideal for food spoilage detection and timely forest fire detection by detecting the gases present and tracking temperature and humidity changes. Its AI features and new BME AI-Studio software help rapidly develop custom solutions for specific use cases. The sensor detects many gases, including volatile organic compounds, volatile sulphur compounds, and other gas types such as carbon monoxide and hydrogen in ppb range. Current consumption can be configured between 2.1µA and 11mA depending on the required data rates and functions. It measures just $3.0 \times 3.0 \times 0.9$ mm³.

Bosch Sensortec
www.bosch-sensortec.com

Circular power connectors

Bulgin 9000 series Buccaneer multi-pin circular power connectors comprises a range of high-amperage flex body, flex in-line, and flange mount connectors for high current applications where easy connection and protection from high levels of dust and moisture are critical. These are available with either a bayonet ¼ turn locking mechanism for quick connector mating and release



or a screw turn locking mechanism that achieves a secure, robust connection but is also easy to disconnect. Both variants have visual alignment features to ensure correct mating, and positive feedback on the locking mechanism. Suitable for mains power to signal applications in multiple industrial environments, the connectors feature robust housings and are available as male and female, with a choice of 2 to 11 poles up to 225A, 1250V AC RMS as standard.

Bulgin
www.bulgin.com

MODULES & BOARDS

High-precision GNSS module

Current general GPS modules are prone to deviation up to hundreds of metres. The RTK-1010 module from LOCOSYS provides robust GNSS services for wireless communication, embedded



systems, avionics, and automotive and consumer electronics. It can concurrently support all satellite positioning systems. RTK-1010 has 135 satellite receiving channels and 65mA ultra-low power consumption. It can also be used as a base station or rover station. The RTK high-precision positioning is more appropriate to unmanned mobile vehi-



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